

Special topics in logic and security II *

June 15, 2023

1. The code C consists of the words (x, y, z) of length 3 over the alphabet $\mathbb{F}_3$ that satisfy $x + 2y = 0$.
 - (a) Make a list of the code-words.
 - (b) How many words of every possible weight do really occur in the code? Write down the weight polynomial.
2. Let C be the code from the previous exercise and let $C^\perp$ be its orthogonal complement.
 - (a) Apply the Theorem of MacWilliams in order to find the weight polynomial of $C^\perp$.
 - (b) List the code-words of $C^\perp$ and verify that the coefficients of the weight polynomial really correspond with the number of words of every given weight.
3. On a table at CERN there are two independent globes. The red globe has the superposition $r = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ while the blue globe has the superposition $b = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$. We interpret by $|1\rangle$ that an event happens in the globe.
 - (a) How probable is to observe an event in the red globe? How probable is to observe an event in the blue globe?
 - (b) Compute the superposition for the composite system. How probable is to observe an event in the composite system?
4. In the same laboratory as in the previous exercise, the state superposition of the two-globe system is $c = \frac{1}{\sqrt{2}}|0\rangle|0\rangle - \frac{1}{\sqrt{2}}|1\rangle|1\rangle$.
 - (a) What is the superposition for the red globe? What is the superposition for the blue globe?
 - (b) How probable is to observe an event in the composite system?
5. Let H_n be a complex Hilbert space. The state $|a_1\rangle \in H_n$ is used as a blank symbol. A unitary function $U : H_n \otimes H_n \rightarrow H_n \otimes H_n$ such that for every state $|x\rangle \in H_n$ one has:

$$U(|x\rangle|a_1\rangle) = |a_1\rangle|x\rangle,$$

is called quantum switch machine. Is a quantum switch machine mathematically possible? Prove your answer!

*Master, 4-th semester. IF